

QUINTAURIS

De la Nube a la FPGA:
Introducción Práctica a
RISC-V con Quintauris
(II ed.)

Angel Berrio
Chief Product Officer





Introducción Curso

Title:

De la Nube a la FPGA: Introducción Práctica a RISC-V con Quintauris (II ed.)

Key Info:

•**Fechas:** 7–17 septiembre 2026

•**Duración:** 21 horas

•**Dirige:** Almudena Rivadeneyra | **Coordina:** Ángel Berrio (Quintauris)

Objetivo:

Obtener una introducción en RISC-V y su ecosistema, con un enfoque teórico-práctico.



Agenda

SEMANA 1

LUN 7
17:00–19:00

1. Fundamentos de RISC-V: de la arquitectura al ecosistema

Quintauris · Ángel Berrio

MAR 8
17:00–19:00

2. Presentación de Quintauris: visión, modelo de negocio y productos

Quintauris · Ángel Berrio

MIÉ 9
17:00–19:00

3. Explorando la plataforma Quintauris, Tools and Infra — incluyendo Testing, Benchmarking y QNTAF (I)

Quintauris · Enrique Pallares

JUE 10
17:00–19:00

4. Explorando la plataforma Quintauris, Tools and Infra — incluyendo Testing, Benchmarking y QNTAF (II)

Quintauris · Enrique Pallares

SEMANA 2

2 Semanas – 8 sesiones

LUN 14
16:30–19:30

5. IP hardware RISC-V open source: introducción, licencias y repositorios

Quintauris · Ignacio Ruiz

MAR 15
16:30–19:30

6. Parte I: De la nube a la FPGA — flujo de despliegue de una IP en la FPGA

UGR & Quintauris · Jose Luis Martínez, Ignacio Ruiz / UGR

MIÉ 16
16:30–19:30

7. Parte II: De la nube a la FPGA — flujo de despliegue de una IP en la FPGA

UGR & Quintauris · Jose Luis Martínez, Ignacio Ruiz / UGR

JUE 17
16:00–20:00

8. Hackathon final: integración de conceptos y solución de un reto práctico

Quintauris · Jose Luis Martínez, con mentores de UGR



Introducción a la arquitectura RISC-V: historia, filosofía y evolución.

Agenda



01 Introduction to RISC-V Architecture

02 Instruction Set

03 Privilege Levels

04 Non-ISA specifications

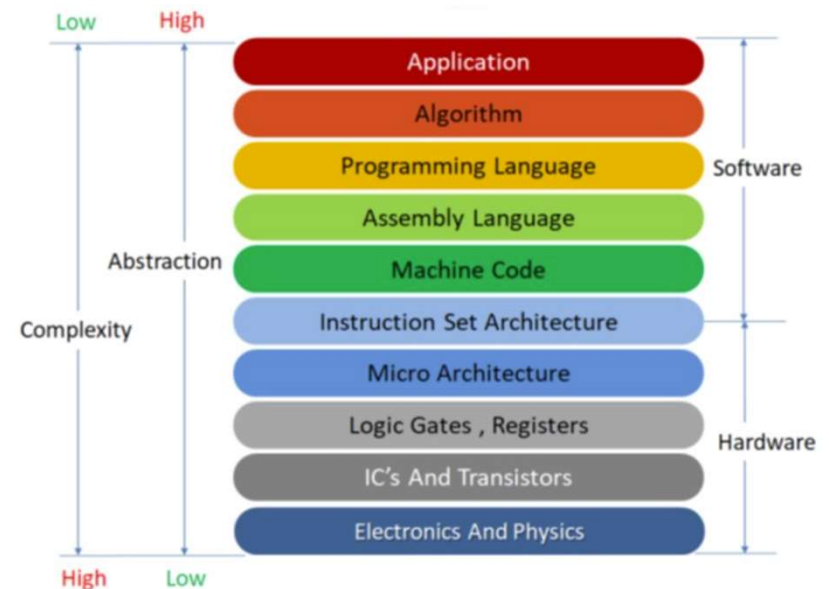
What is an Instruction Set Architecture (ISA)?



An Instruction Set Architecture (ISA) is the complete specification of how software interacts with computer hardware. It serves as the crucial interface between hardware and software, defining:

- The set of instructions a processor can execute
- Registers, memory addressing modes, and data types
- Exception handling mechanisms
- Memory architecture and I/O operations

The ISA is essentially the "contract" that hardware designers must implement, and software developers must adhere to, ensuring compatibility across different implementations.



RISC VS CISC: Technical comparison



Feature	RISC	CISC
<i>Instruction Set Size</i>	Small, simplified	Large, complex
<i>Number of Instructions</i>	Fewer instructions	Many instructions
<i>Instruction Complexity</i>	Simple instructions	Complex instructions
<i>Code Size</i>	Larger (more instructions required)	Smaller (fewer instructions required)
<i>Power Consumption</i>	Lower	Higher
<i>Typical Examples</i>	ARM, MIPS, PowerPC	Intel, AMD -> x86/x64
<i>Performance Strategy</i>	Simplified hardware, faster instruction execution	Powerful instructions, fewer instructions per program

RISC VS CISC: Technical comparison



Incrementing a variable: Same problem, different solutions

NXP s08 (CISC)	ARM Cortex M (RISC)
INC counter	LDR R0, =counter LDR R1, [R0] ADD R1, #1 STR R1, [R0]

RISC VS CISC: Market adoption and projection



Aspect	RISC	CISC
<i>Current Market Presence</i>	Mobile, IoT, wearables, embedded systems	PCs, laptops, servers
<i>Key Industry Examples</i>	ARM(Qualcomm, Samsung)	Intel (Core), AMD (Ryzen)
<i>Growth Drivers</i>	Power efficiency, simplified design, custom applications	Legacy software, powerful computing capabilities
<i>Future Projection (5-10 yrs)</i>	Rapid growth, expanding to servers, laptops, automotive, HPC	Limited growth

The RISC Philosophy



Simple Design

Fixed-length instruction and simple addressing modes -> Less complex HW & compiler.

Load-Store Arch.

Memory access only through specific load/store instructions, with all other operations using registers -> clean execution pipeline.

Large Register Files

More general-purpose registers reduce memory access, alleviating the "von Neumann bottleneck" between processor and memory.

The RISC (Reduced Instruction Set Computer) philosophy embraces "simple instructions, efficient execution" as its core principle. By eliminating complex instructions that require multiple clock cycles, RISC designs achieve higher performance through pipelining and optimization of frequently used instructions.

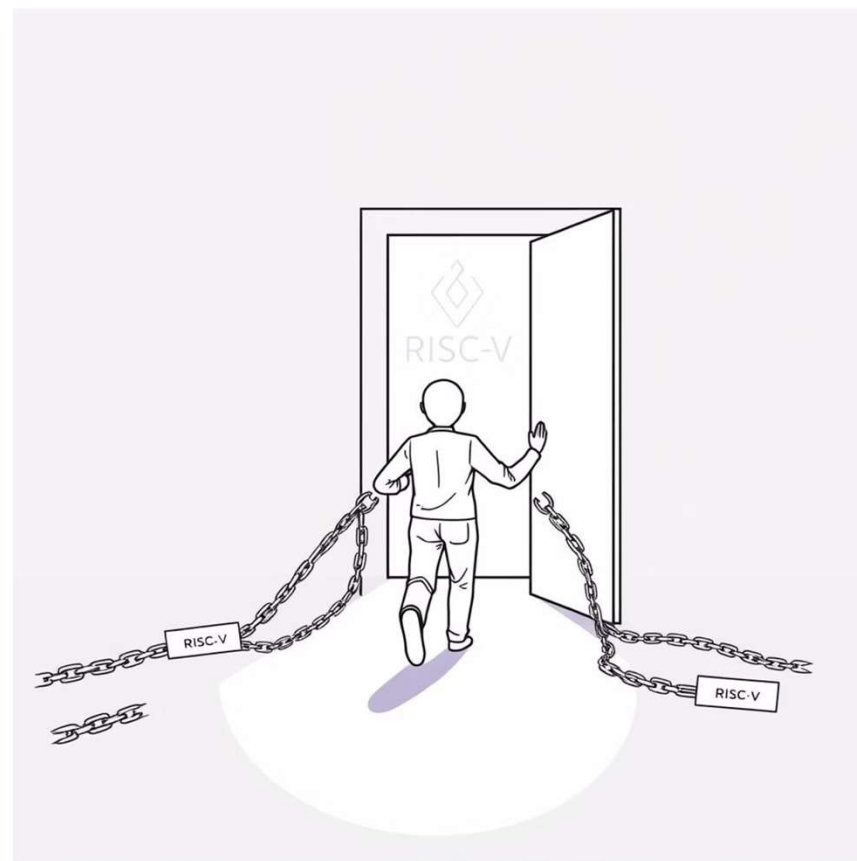
Why RISC-V?

The Need for an Open ISA

Proprietary ISAs like x86 (Intel/AMD) and ARM challenges:

- Licensing costs
- Restrictions on customization and innovation
- Geopolitical vulnerabilities in the supply chain
- Difficulty in teaching processor design with closed architectures

RISC-V addresses these issues through its open-source, royalty-free model that enables unrestricted development.



"RISC-V is to processor design what Linux was to operating systems."

Birth of RISC-V: From Berkeley to the World



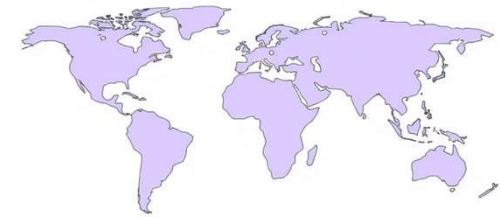
Academic Origins

Developed at UC Berkeley by Krste Asanović, Andrew Waterman, and Yunsup Lee to address the limitations of existing ISAs for research and teaching.



Open License

Released under permissive BSD license, allowing unrestricted use in both academic and commercial applications without royalty payments.



Global Expansion

Transition from university project to international standard managed by RISC-V International, with over 3,000 members across 70 countries.

RISC-V Today: Ecosystem and Adoption



The RISC-V ecosystem has seen explosive growth across multiple domains:



Hardware

SiFive, Microchip, Espressif, and Western Digital shipping commercial RISC-V products



Software

Linux, Android, FreeRTOS support; LLVM and GCC compiler toolchains



Education

Incorporated into computer architecture curricula worldwide

Major tech companies including Google, Qualcomm, NVIDIA, and Alibaba have joined the RISC-V foundation and are developing RISC-V solutions for diverse applications from IoT to high-performance computing.



In 2022, the European Processor Initiative chose RISC-V as the foundation for Europe's exascale computing initiative, signaling its growing importance in strategic technology independence.

Key Technical Features of RISC-V

Modular Design

RISC-V follows a base-and-extension model, with a minimal required instruction set (RV32I/RV64I) and optional standard extensions (M for multiplication, F/D for floating-point, etc.).

Scalability

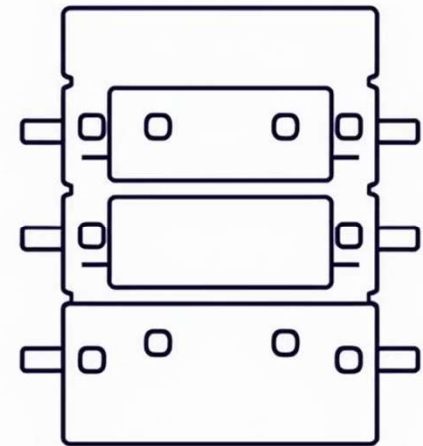
Implementations range from tiny microcontrollers (RV32E) to high-performance server processors (RV64GC), all sharing the same fundamental architecture.

Custom Extensions

Reserved instruction space for vendor-specific extensions, enabling domain-specific optimizations for AI, cryptography, or signal processing without breaking compatibility.

Future-Proof Design

Clean separation between privileged and unprivileged instructions allows for virtualization, security features, and advanced memory management without ISA revision.



RISC-V ISA Specification Structure

1

Volume I: User-Level ISA

Defines base instructions accessible to all software (Unprivileged Spec)

2

Volume II: Privileged Architecture

Defines privilege levels, control/status registers, and system-level features

3

Extensions

Add capabilities like compressed instructions, floating point operations, and virtualisation

The modular structure allows implementers to select only the features needed for their specific applications, creating a highly customisable architecture.

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01 Introduction to RISC-V Architecture

02 Instruction Set

03 Privilege Levels

04 Non-ISA specifications

Modular Design of RISC-V



Minimal Base Set

Starts with a minimal base set of instructions.

- RISC-V's modular design offers flexibility and customization.
- The base instruction set provides core processor functionality.
- Additional instruction sets extend capabilities for specific applications.



Customizable Extensions

Allows for extensions to customize for specific applications.

Base Integer Instruction Sets



- RV32I and RV64I are the base integer instruction sets.
- Provide fundamental instructions for arithmetic and logic operations.
Less than 50 HW instructions.
- Ratified 2019 supported forever after.

RV32I

32-bit instruction set.

RV64I

64-bit extension of RV32I.

RV32E

16-register subset of RV32I

RV32I



Basic functionality with 32-bit integers (its 64-bit variant is RV64I).

Includes instructions for:

- Addition
- Subtraction
- Bitwise ops
- Load and store
- Jumps
- Branches



Register	ABI Name	Description
x0	zero	Hard-wired zero
x1	ra	Return address
x2	sp	Stack pointer
x3	gp	Global pointer
x4	tp	Thread pointer
x5	t0	Temporary/alternate link register
x6-7	t1-2	Temporaries
x8	s0/fp	Saved register/frame pointer
x9	s1	Saved register
x10-11	a0-1	Function arguments/return values
x12-17	a2-7	Function arguments
x18-27	s2-11	Saved registers
x28-31	t3-6	Temporaries

The base ISA also specifies the 32 CPU registers, which are all 32-bits wide, plus the program counter. The only special register is x0, which always reads 0, as implemented in many previous RISC ISAs.



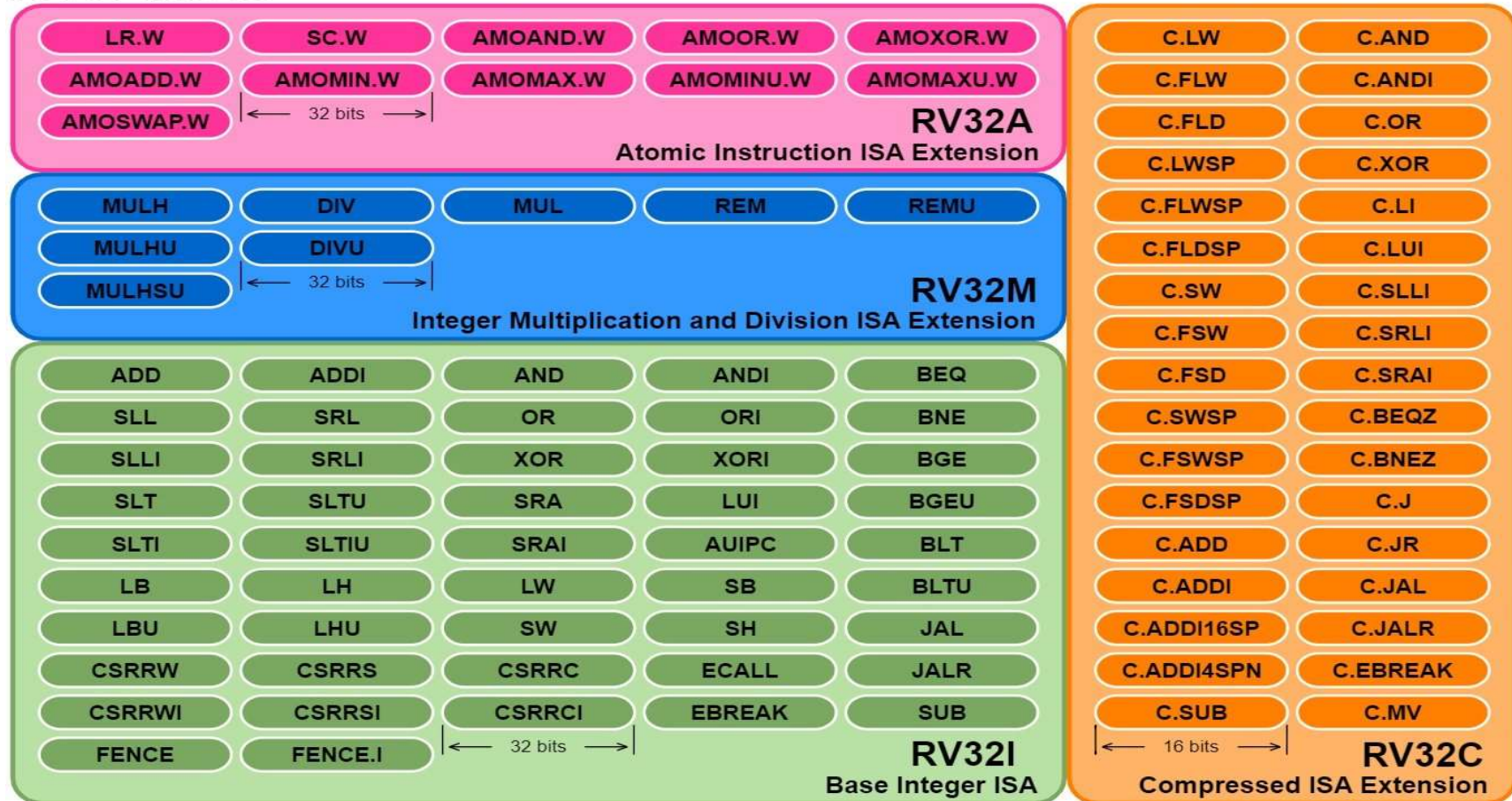
RISC-V Instruction Set Extensions

Extensions add functionality to the base instruction sets.

Tailored for specific applications and performance requirements.

Extension	Description
M	Multiplication and Division
A	Atomic Operations
F	Single-Precision Floating-Point
D	Double-Precision Floating-Point
C	Compressed Instructions

RV32IMAC



M Extension: Integer Multiplication and Division



Core Operations

Adds essential instructions for multiplication (mul), division (div) and remainder (rem) operations to the base ISA.

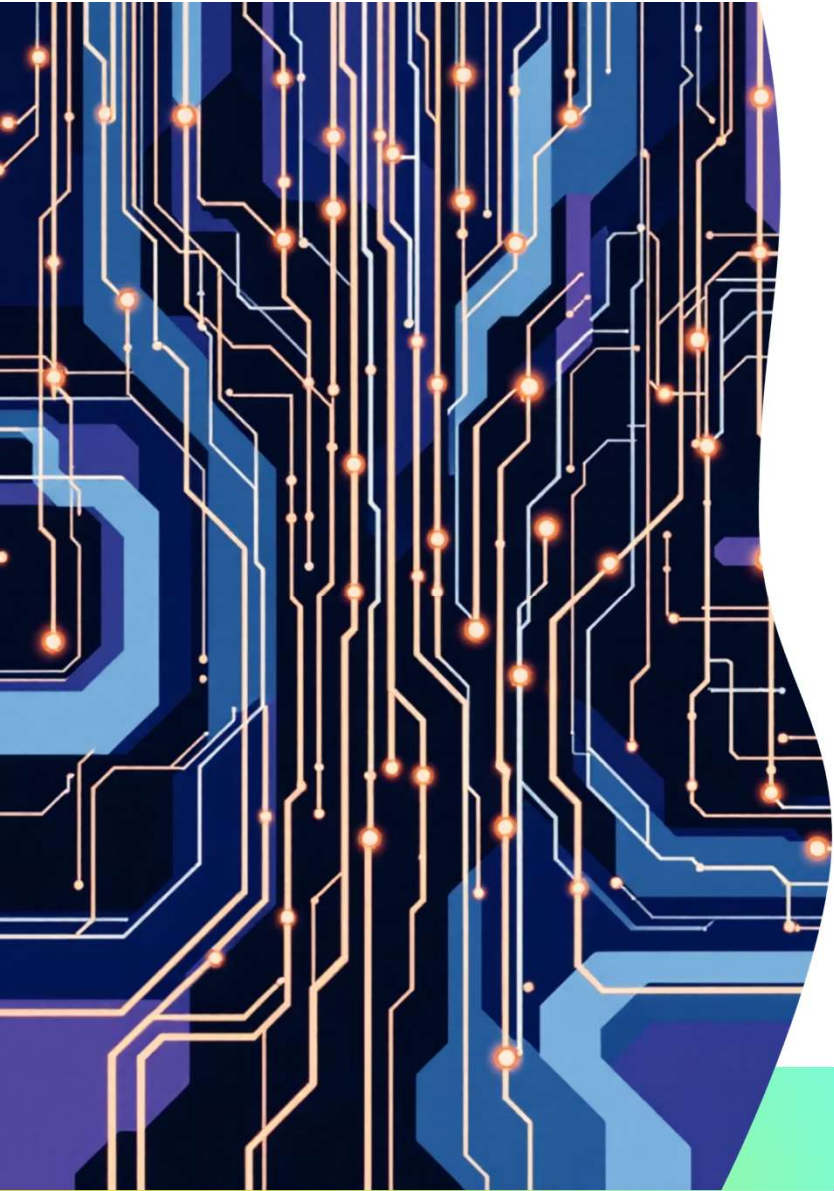
Performance Impact

Dramatically improves computational efficiency for mathematical operations that would otherwise require multiple instructions and cycles.

Maturity

One of the most established extensions, ratified and stable since early ISA versions and widely implemented across RISC-V processors.

The M extension is considered essential for most practical implementations, forming part of the common RV32GC and RV64GC profiles.



Vector Extension (V): Large-Scale Parallel Computing



1

Vector Registers

Supports vector registers of 128+ bits ($VLEN \geq 128$), enabling significant data parallelism

2

SIMD Operations

Enables Single Instruction Multiple Data operations critical for HPC, AI and multimedia processing

3

Flexible Architecture

Variable vector length and register grouping (LMUL) adapt to different workloads

The Barcelona Supercomputing Center is pioneering research with this extension, demonstrating its potential for scientific computing.

Real-World Use Cases and Ecosystem

Data Centre & HPC

Ventana leading RISC-V adoption with AI and high-performance computing applications leveraging vector extensions

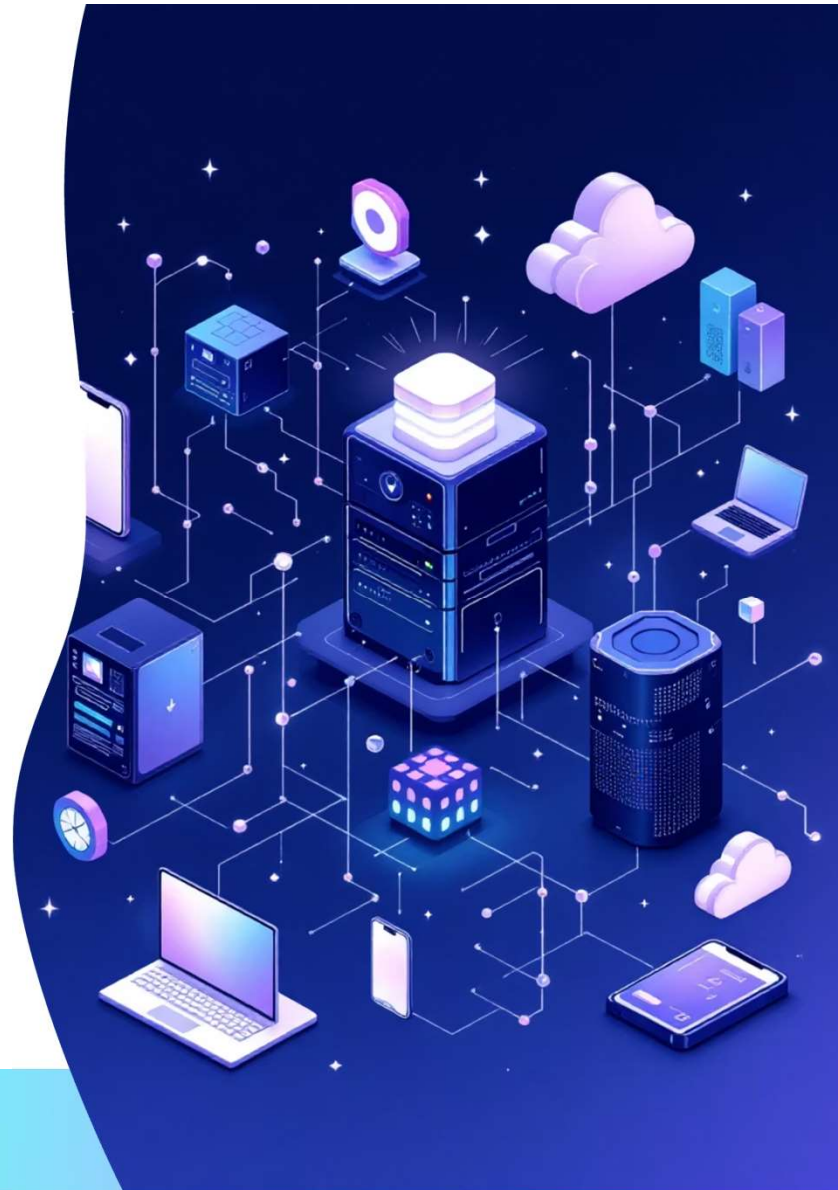
Mobile SoCs

Mobile processors integrating RISC-V cores with security and multimedia extensions for specific subsystems

Community Growth

Active community sharing developments at events like RISC-V Summit Europe 2025

The flexibility of RISC-V extensions is enabling a new generation of specialised processing solutions across the computing spectrum, from embedded to high-performance applications.



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The Three Privilege Levels in RISC-V



User (U-mode)

The lowest privilege level intended for user applications. Provides limited access to system resources for safety and security. Operations are restricted to prevent system tampering.

Supervisor (S-mode)

Optional intermediate level designed for operating systems. Manages memory resources, handles system calls, and provides abstraction for applications. Enables virtual memory implementation.

Machine (M-mode)

The highest privilege level, mandatory in all RISC-V hardware. Has complete control over the processor and memory. Manages the most critical security-related tasks.

Common configurations include M-mode only; M+U modes; or the most robust M+S+U mode combination for sophisticated systems.

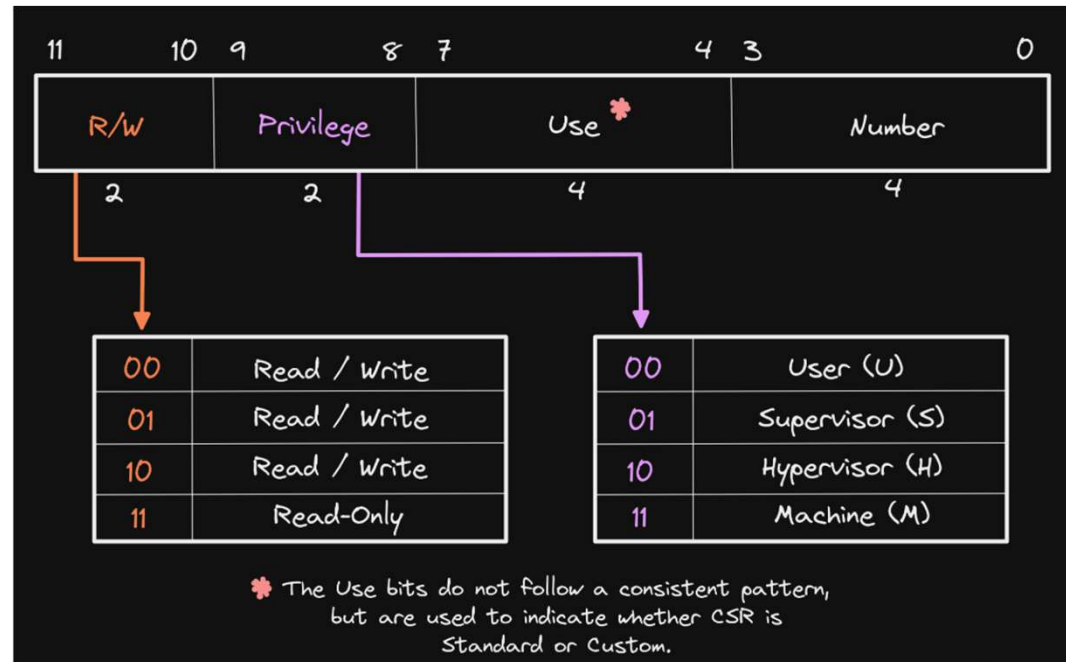
Control and Status Registers (CSRs)



Special registers defined in the Privileged Spec that provide critical system control capabilities:

- Interrupt configuration and handling
- Exception management
- Memory protection and translation
- Timer and performance counters
- Privilege transitions

Access to CSRs is restricted by privilege level to enforce security and isolation between software components.



Key Control and Status Registers (CSRs)

Register	Function	Access Level
<code>mstatus</code>	Controls global state, privileges, and enables interrupts	M-mode
<code>mtvec</code>	Base address for interrupt vectors in M-mode	M-mode
<code>medeleg</code>	Exception delegation to S-mode	M-mode
<code>mideleg</code>	Interrupt delegation to S-mode	M-mode
<code>mhartid</code>	Hardware thread (hart) identifier	M-mode

These registers form the critical control plane allowing higher privilege modes to configure and monitor lower privilege execution, establishing the foundation for the security architecture.

Privilege Transitions and Exceptions



Privilege Boundaries

Hardware enforces strict boundaries; illegal operations cause traps that transfer control to higher privilege levels



Return Instructions

MRET and SRET instructions return from exceptions, carefully restoring previous privilege state



Delegation Mechanisms

Configurable delegation allows lower privilege levels to handle specific interrupts/exceptions efficiently

This architecture creates a secure framework for multi-layered software stacks whilst minimising performance overhead.

Hypervisor Extension and Virtualisation



H-Extension

Adds hypervisor capabilities to the RISC-V architecture, enabling sophisticated virtualisation support.

Efficiency

Enables near-native performance for virtualised systems through hardware-assisted virtualisation.



Virtual Levels

Creates virtualised privilege levels (HS, VS, VU) for running multiple isolated operating systems.

Server Applications

Critical for server, cloud and automotive environments where multiple workloads must coexist securely on shared hardware.

The H-extension transforms RISC-V into a competitive platform for data center, automotive and cloud applications by enabling robust hardware-based virtualisation with minimal performance overhead.

Practical Applications and Examples



Simple Embedded Systems

Utilise M-mode only for maximum efficiency in resource-constrained environments:

- Microcontrollers for IoT devices
- Real-time control systems
- Wearable technology

Unix-like Operating Systems

Implement full M+S+U stack for comprehensive security and multitasking:

- Desktop and laptop computers
- Mobile devices
- Single-tenant servers

Advanced Virtualisation

Leverage H-extension for isolation in multi-tenant environments:

- Cloud infrastructure
- Consolidated server environments
- Secure multi-domain systems

RISC-V's privilege architecture scales elegantly from the simplest embedded devices to the most complex server infrastructure, providing appropriate security mechanisms for each context.

From SW perspective

By default, the processor starts in M-mode

The **bootloader** or **OS** change the mode at start up:

1. Set the return program counter (*mepc*) with the address where user code will begin.
2. Change the MPP field (Machine Previous Privilege) in *mstatus* to 00 (U-mode).
3. Execute the *mret* instruction (machine return).
 - This loads *mepc* into the PC.
 - Execution mode changes to U.

Setup in M-mode

```
la t0, user_start      # 1. Address of user program
csrw mepc, t0
```

```
li t1, 0
csrrs x0, mstatus, t1  # 2. clear MPP bits
```

```
mret                    # 3. Jump to user_start in U-mode
```


Privileged Modes day II

From HW perspective:

Case: a program in **U-mode** tries to execute a privileged instruction (*csrw mtvec, t0*)

a) Status register (mstatus)

Contains the **current mode**: 2 bits indicate 00=U, 01=S, 11=M.

These bits travel through the pipeline as a **global privilege signal**.

b) Decode unit

The decoder identifies if the instruction is **privileged** (e.g., *csrw mtvec*).

It generates a signal: `instr_privileged = 1`.

c) Privilege check logic

A logic gate compares `allow=(priv_level >= required_level)`

If `allow=0`, then `trap_signal=1` is asserted.

d) Trap control unit

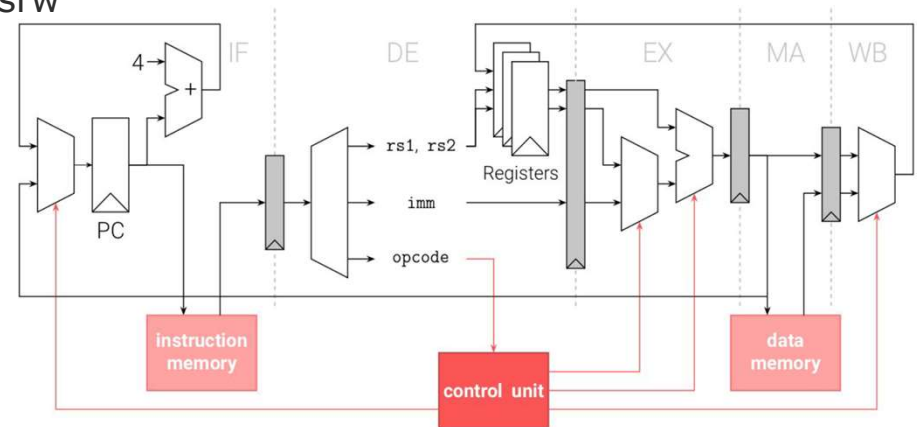
When `trap_signal=1`:

- The pipeline is flushed.

- The PC is loaded from *mtvec*.

- The old PC is saved into *mepc*.

- mcause* is updated with the reason for the exception.



This is literally implemented with AND/OR gates and multiplexers.

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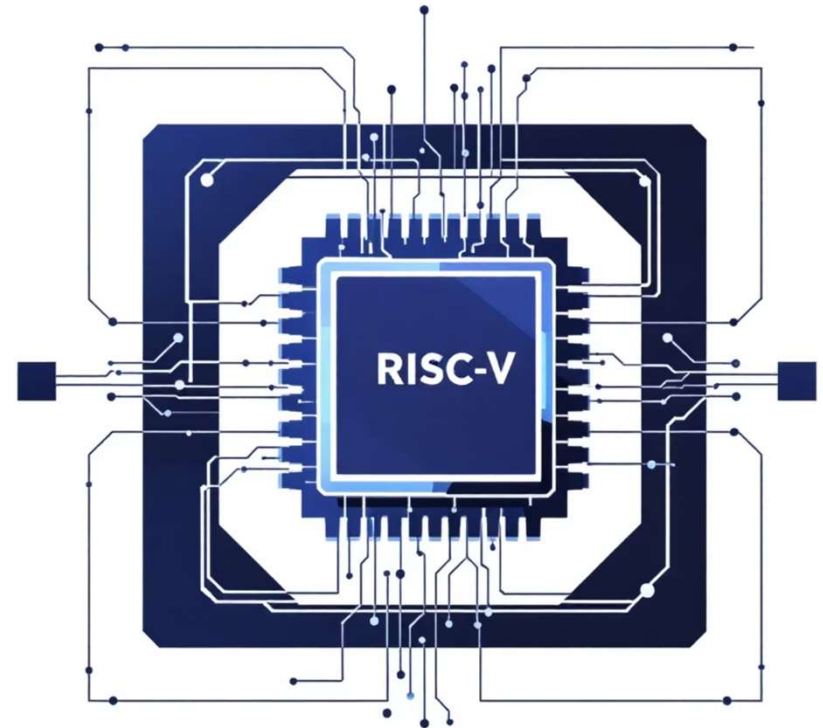
04 Non-ISA specifications

What Are Non-ISA Specifications?



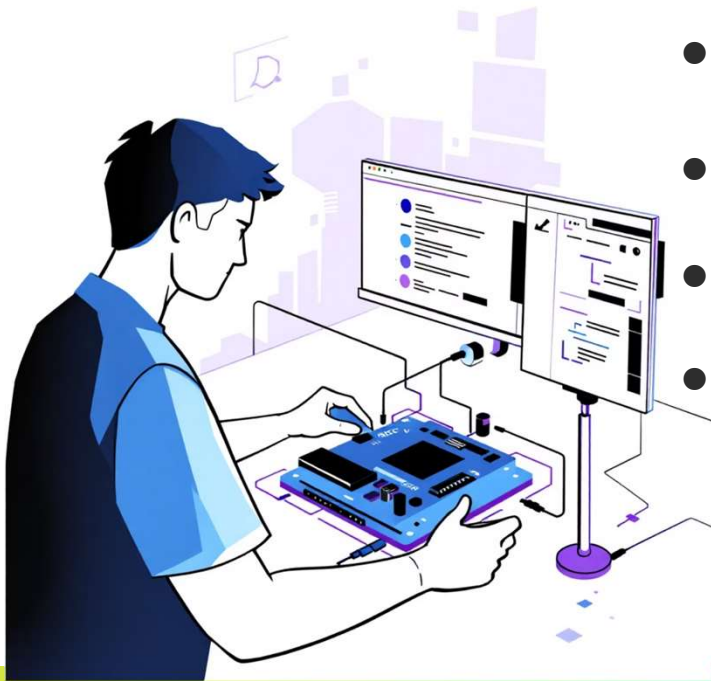
Elements that do not add/modify instructions or opcodes

Define hardware interfaces, debug systems, and platform standards



RISC-V Debug Specification

v1.0, February 2025



- Flexible architecture for diverse platforms
- Breakpoints, watchpoints, debug registers
- Tool interoperability across cores/SoCs
- Critical for development & field diagnostics

Quality of Service Register Interface

v1.0, June 2024



Identify Workloads

Tag access types to shared resources



Allocate Resources

Capacity and bandwidth management



Performance Isolation

Differentiated request handling

Profile



Profile

A RISC-V Profile is a set of features and extensions to ensure compatibility and standardization across different RISC-V implementations. These profiles specify a minimum set of capabilities that compliant processors must implement, enabling software developers to write applications that can run on any RISC-V processor adhering to a specific profile.

Bases	Release 1		Release 2		Future	
RV32I RV64I	RVI[20] RV120U32 RV120U64 RV32I RV64I	RVA20 RVA20U64 RVA20S64 RV64I	RVA22 RVA22U64 RVA22S64 RV64I	RVA23 RV64I		
Load Store Jumps Branches Add Subtract Logical	On years released, this only has a Mandatory Base All other compatible ratified extensions are optional	Mul/Div Atomics Compressed Float Double Priv 1.11 MemRegions Fences VirtualMem	Vector Bitmanip Scalar Crypto FP16 Priv 1.12 Hypervisor Cache	Vector Crypto PtrMasking BFloat16 Zcompressed Priv 1.13	Android Features	More profile types: RVB, RVM RV128 Matrix Ops SPMP/IOPMP CFI CHERI GPU 48/64 bit instructions
	MAJOR		MAJOR			

- Only a subset of extensions are listed above and it is not an exhaustive list
- Some extensions may be optional or non-profile in one profile and be mandatory in another

Key Components of a RISC-V Profile

Base ISA

Each profile specifies a base instruction set, which includes the fundamental instructions that all compliant processors must support.

Standard Extensions

Profiles include standard extensions that add specific functionalities. For example, the "M" extension for integer multiplication and division or the "F" extension for single-precision floating-point operations.

Privileged Architecture

The profiles may also define a privileged architecture to ensure that the processor can support operating systems and other system-level software.

Optional Extensions

Optional extensions could be included, as well, for additional functionalities.

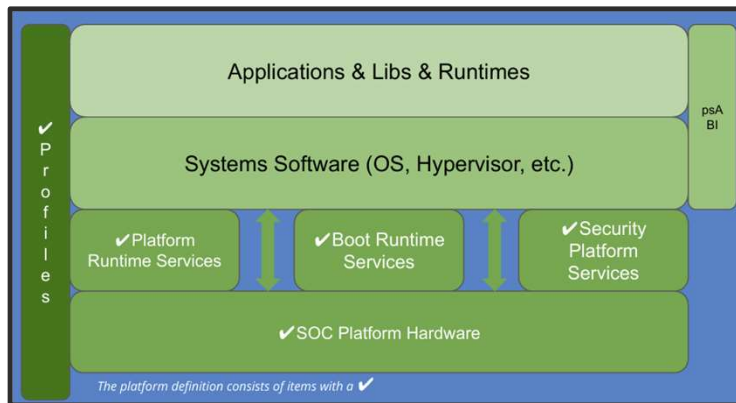
Platform



Platform

A RISC-V Platform defines the necessary firmware and hardware for a RISC-V platform to run various operating systems.

The goal of the platform specification is to ensure interoperability and consistency across different RISC-V implementations, providing a standardized environment for operating systems and software developers.



Key Components of RISC-V Platform

Profile

RISC-V profile definition.

Boot Runtime Services: To provide the necessary functions during the system boot process before the control is handed over to the operating system.

Platform Runtime Services:

Runtime Services are available to the operating system after the boot process is complete. Including variable, time, virtual memory and reset services.

SOC Platform Hardware:

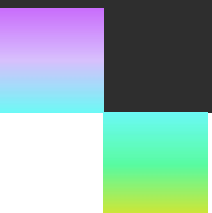
The RISC-V platform definition for SOC (System on Chip) hardware encompasses a broad range of components and features, ensuring a flexible and scalable environment for development and integration.

Server Platform Specification

Accelerating RISC-V in the data center



- Standard hardware/software capabilities
- Portable system software across servers
- Cross-implementation interoperability
- Evolving to accelerate data centre adoption



The RISC-V Ecosystem: Tools, Communities, and Use Cases.

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01 Introduction to RISC-V Ecosystem

02 RISC-V International

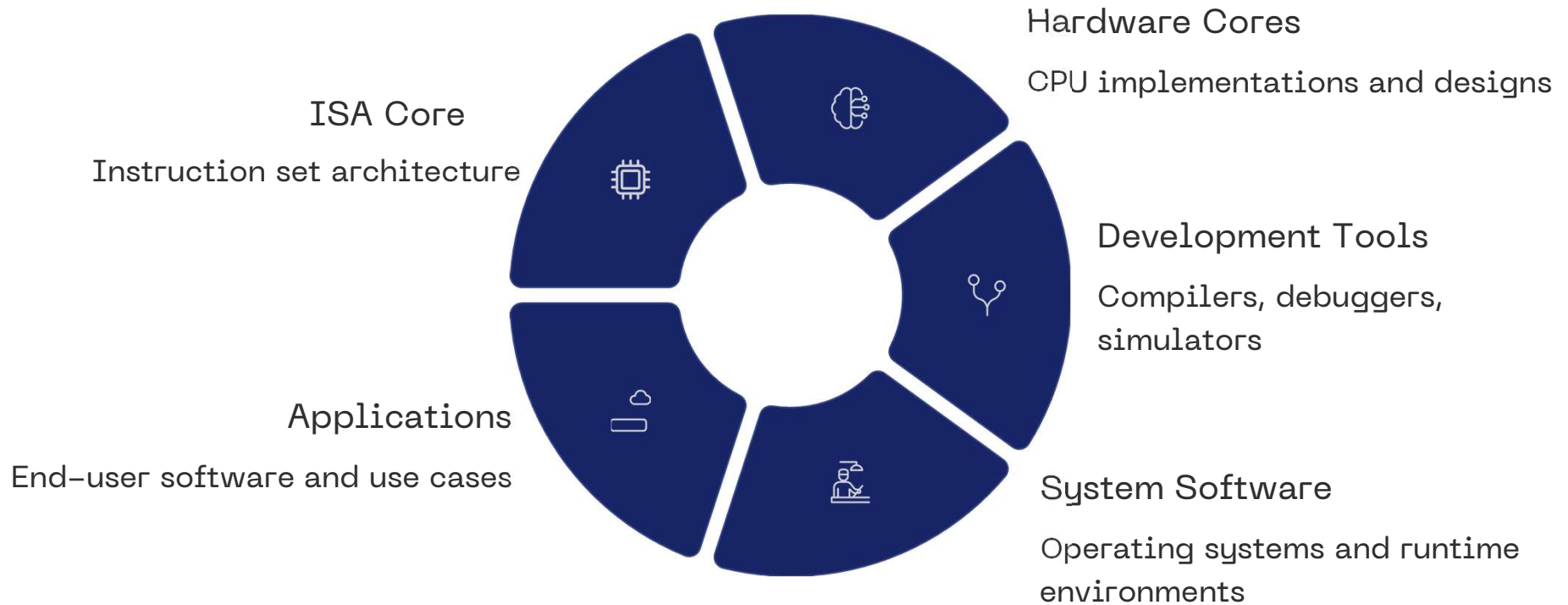
03 RISE

04 RISC-V Market Status

The RISC-V Ecosystem



Beyond the ISA: RISC-V represents a complete ecosystem encompassing hardware, software, people, and standards that work together to enable innovation.



RISC-V Development Stack

Complete software infrastructure for embedded development



Applications

User software and embedded applications



Operating Systems

Linux, FreeRTOS, bare metal environments



Hardware Frameworks

Chisel, SystemVerilog, FPGA tools



Debug & Simulation

GDB, Spike, QEMU, Renode



Toolchain

GCC, LLVM, compilers, linkers



Each layer builds upon the foundation below, creating a comprehensive development environment that rivals established architectures.



Compilers & Toolchains



GCC Toolchain

Complete support for RV32 and RV64 architectures with extensive optimization capabilities and mature ecosystem integration.

- Full C/C++ standard library support
- Robust optimization passes
- Newlib and glibc compatibility



LLVM/Clang

Modern compiler infrastructure with growing RISC-V support and excellent IDE integration for development workflows.

- Advanced static analysis
- IDE integration (VS Code, CLion)
- Modular architecture

Cross-compilation essentials enable development on x86 hosts while targeting RISC-V embedded systems efficiently.

Open-Source Hardware Cores

Community-driven implementations powering research and commercial products

Rocket Chip (CHIPS Alliance)

Berkeley's scalable generator framework supporting in-order execution with comprehensive system integration capabilities.

- Configurable pipeline stages
- Academic proven

Sargantana (BSC)

In-Order RISC-V Processor

- Vector Extension
- 1 GHz+

Hazard3

Minimal area-optimized 32-bit core perfect getting started with small footprint design.

- Ultra-compact design
- RaspberryPi Pico 2
- Educational friendly

CVA6 (OpenHW)

64-bit application-class processor with MMU support, suitable for running Linux and complex operating systems.

- MMU support
- Linux capable





Commercial Hardware Cores



Production-ready implementations with enterprise support and optimization



SiFive

Industry-leading commercial RISC-V cores with comprehensive IP portfolio, design services, and proven silicon implementations across multiple process nodes.



Andes Technology

Extensive portfolio of 32/64-bit cores with custom extensions, advanced debug features, and comprehensive development tools for embedded applications.



Synopsys ARC-V/ MIPS/Global Foundries

Configurable, power- and area-efficient RISC-V processor IP cores. It extends Synopsys's long-standing ARC line.

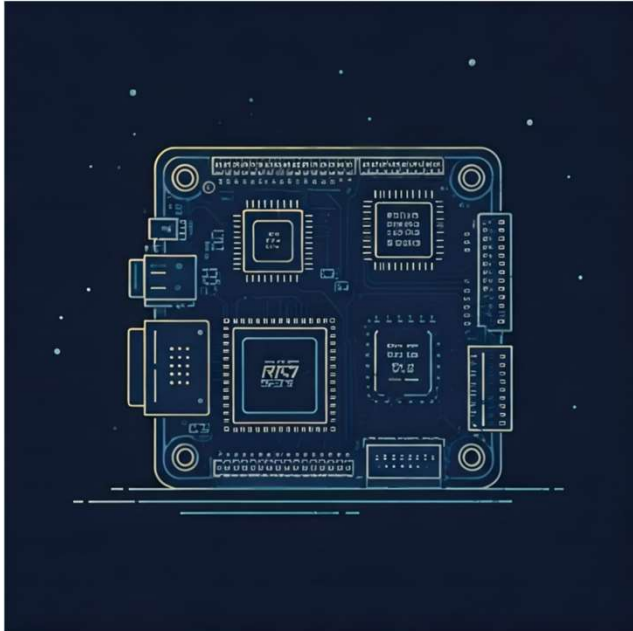
Commercial cores provide enterprise-grade support, extensive validation, and professional services for production deployments.

Hardware Design Tools

Modern approaches to RISC-V processor and system design

Chisel HDL

Scala-based hardware description language enabling high-level design with powerful generators and parameterization for complex processor architectures.



Traditional HDLs

Established Verilog and VHDL design flows with mature toolchain support, extensive IP libraries, and proven methodologies for ASIC development.

FPGA Prototyping

Rapid prototyping platforms enabling hardware validation, software development, and system integration testing before silicon implementation.

The combination of modern HDLs like Chisel with traditional flows and FPGA prototyping creates a comprehensive design environment that accelerates RISC-V development cycles.

What is RISC-V International?

A **non-profit organisation** established to support and advance the open, royalty-free RISC-V instruction set architecture (ISA)

Our mission is to enable global collaboration, foster innovation, and drive ecosystem growth across the semiconductor and technology industries

Governance

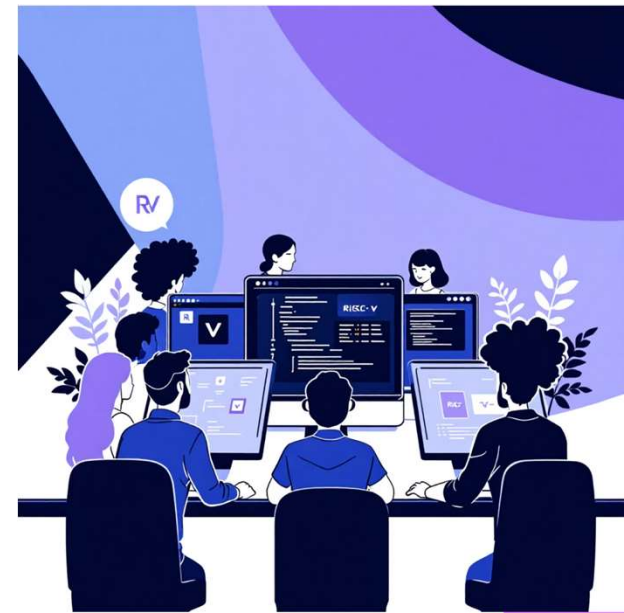
Non-profit organisation with over 3,100 members globally

- Technical Steering Committee oversees ISA development
- Board of Directors includes industry leaders
- Transparent specification development process

Events & Engagement

Active community participation channels

- RISC-V Summit: Annual global conference
- Regional workshops and hackathons
- Working groups with open participation



Membership Tiers: Who Can Join and Why?

Community Members

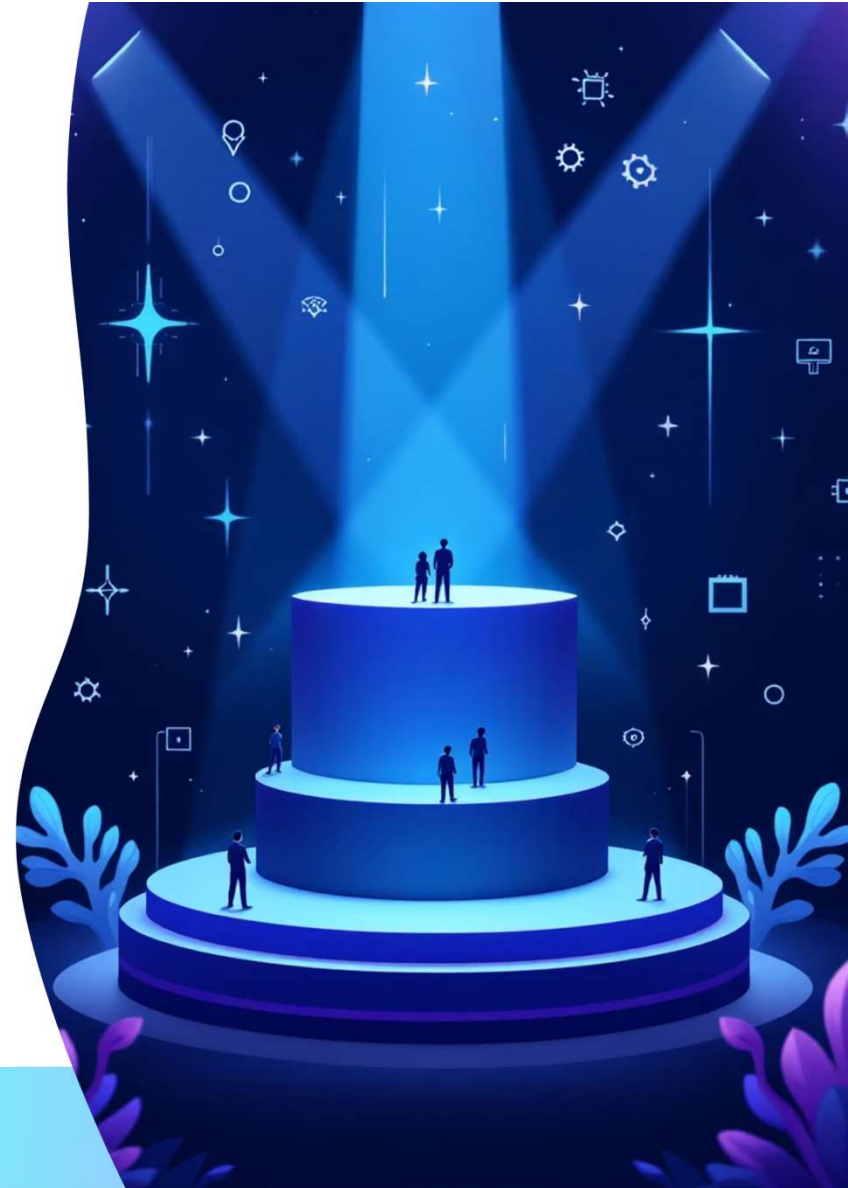
- Free for academics, nonprofits, individuals
- Elect 2 Directors and 1 TSC Delegate
- Access to open specifications

Strategic Members

- Commercial organisations driving adoption
- Elect 3 Directors and 2 TSC Delegates
- Training discounts and trademark use

Premier Members

- Highest tier with expanded roles
- Board Director positions
- Shape RISC-V's future direction



RVI Premier Members

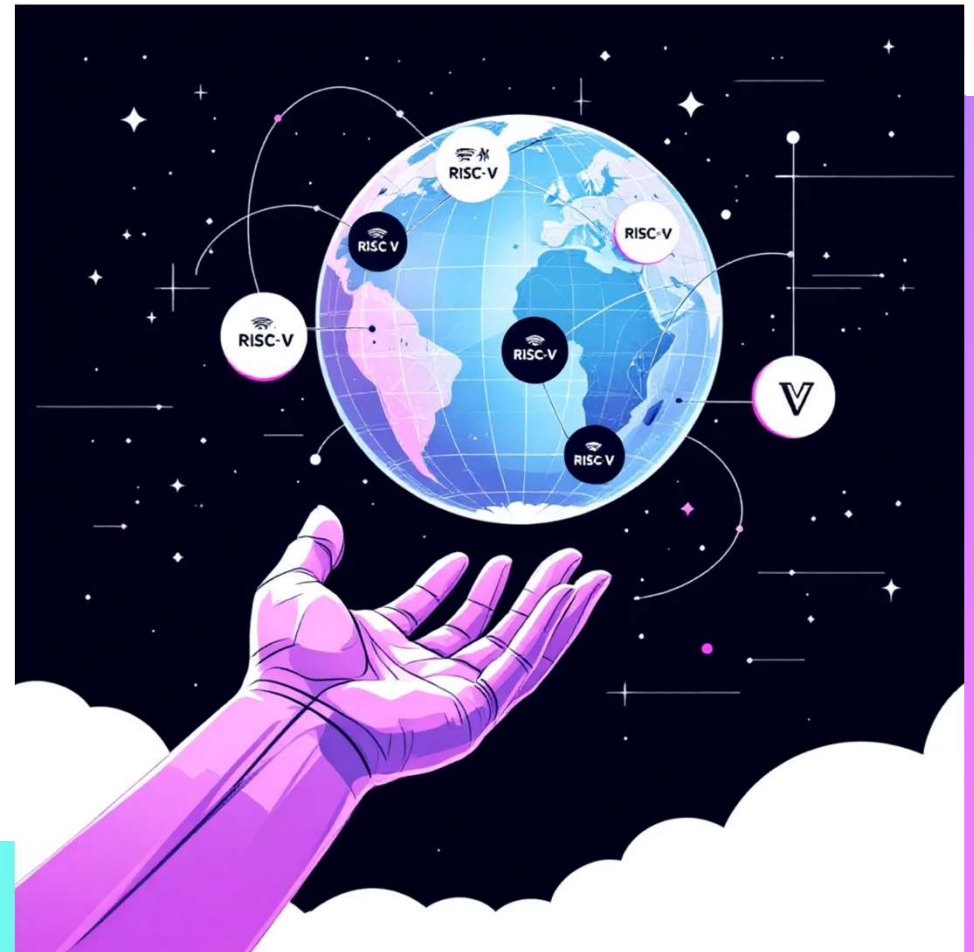


Join RISC-V International: Shape the Future of Open ISA

Become part of a vibrant global community driving semiconductor innovation through open collaboration and shared resources

Influence governance, standards, and training programmes to fit your organisation's strategic needs and technological direction

**Visit riscv.org/members/join
to explore membership options**



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04 RISC-V Market Status

Collaboration: RISE (RISC-V Software Ecosystem)



Raise

RISC-V platform quality



Ensure

RISC-V is tier 1 for tools and libraries



Align

Ecosystem partners efforts for adoption

How: Working Upstream, Transparently

Open and Transparent Organization

Coordinating contributions

Bringing open source communities together

Mobilizing Software Ecosystem for Growth



RISE Members

Premier Members



General Members



Additional Information: <https://riseproject.dev/>

Developer Infrastructure

Build Farm

- Integration to Kernel and GCC CI: more testing to improve quality
- Increase quality while landing large autovectorization changes in GCC with pre-commit and post-commit automated testing
 - 137 bugs found and fixed with fuzzing
 - 792 post-commit builds, 400+ patches tested pre-commit
 - Fixing compiler bugs affecting SPEC

Board Farm

- Some usage of Scaleway EM-RV1
- Partnership with Eclipse Adoptium
 - Released Java 17, 21, 23 and later



Agenda



01 Introduction to RISC-V Ecosystem

02 RISC-V International

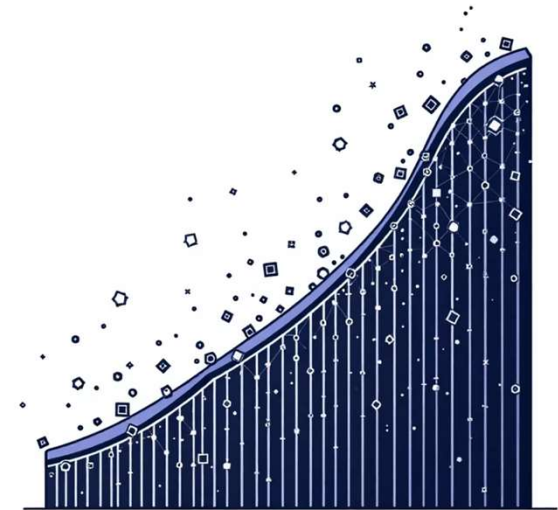
03 RISE

04 RISC-V Market Status

The RISC-V Market Explosion

The RISC-V ecosystem is experiencing unprecedented growth, driven by the industry's shift towards customizable open architectures and specialized computing needs:

- Market valued at **USD 1.76 billion** in 2024
- Projected to reach **USD 8.57 billion** by 2030
- Impressive **CAGR of 30.7%** over the forecast period
- More than **20 billion RISC-V cores** expected in use globally by end of 2026



Infineon's Strategic Moves in RISC-V

"Our focus is on building a robust ecosystem that enables RISC-V to excel in safety-critical applications where real-time performance is non-negotiable."

— Thomas Schneid, Head of Software Partnership, Infineon (RISC-V Summit China 2025)

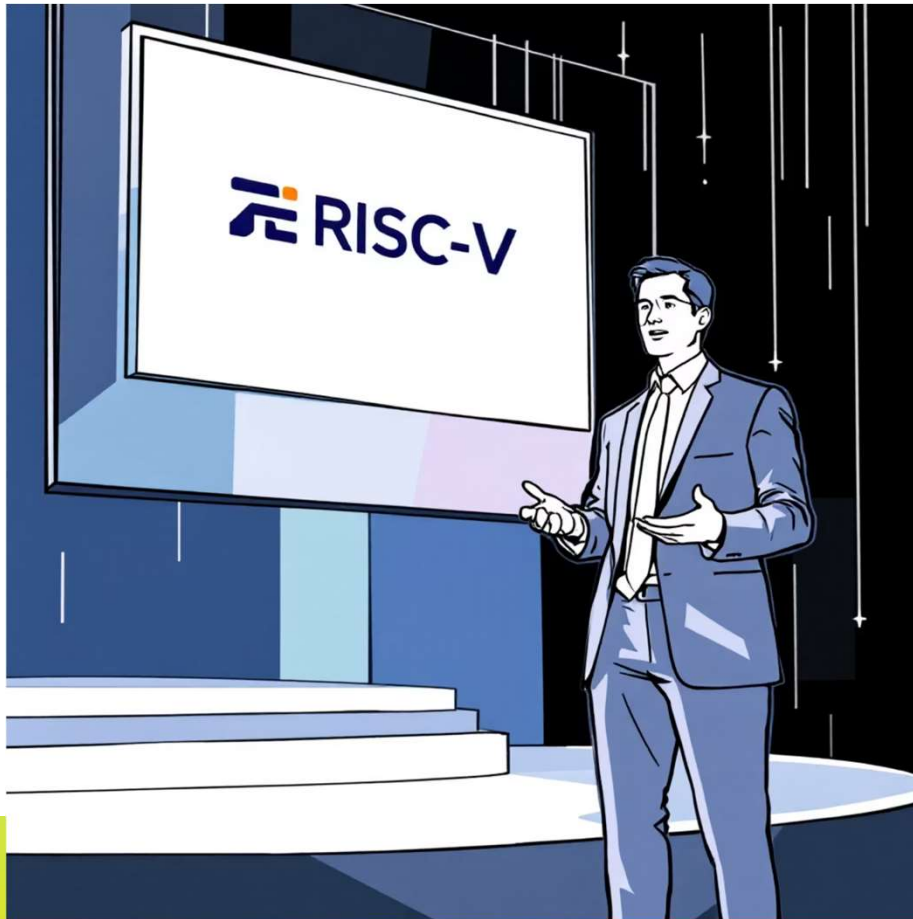
Key Focus Areas

- Automotive safety systems
- Real-time control applications
- Industrial automation

Strategic Initiatives

- Enhanced software tooling
- Cross-industry collaborations
- Ecosystem maturity investments

Qualcomm's Ecosystem Mindset



"RISC-V is far more than just an ISA—it's an ecosystem that requires community-wide collaboration to scale beyond hardware IP."

— Leendert van Doorn, SVP, Qualcomm

- Qualcomm investing heavily in platform-level support
- Targeting mobile and edge AI applications
- Building developer tools and middleware solutions



Nvidia's Game-Changer: CUDA Comes to RISC-V



July 2025 Announcement

Nvidia officially begins porting its CUDA ecosystem to RISC-V CPUs, marking a significant shift in the high-performance computing landscape

Direct Competition

This move enables RISC-V to compete directly with established architectures like Arm and x86 in the lucrative high-performance computing market

Chinese Market Opportunity

Creates a rare opportunity for RISC-V CPU vendors, especially in China, to enter GPU-accelerated markets despite trade restrictions

Challenges on the Road Ahead



Software Ecosystem Maturity

Limited availability of optimised compilers, libraries, and development tools compared to x86 and Arm ecosystems



Fragmentation Risk

Vendor-specific ISA extensions threaten interoperability and could fragment the ecosystem if not properly standardised



Collaboration Needs

Greater coordination required between hardware vendors, software developers, and standardisation bodies

 Without addressing these challenges, RISC-V risks hitting a growth ceiling that could limit its potential in high-value markets despite strong hardware innovation.

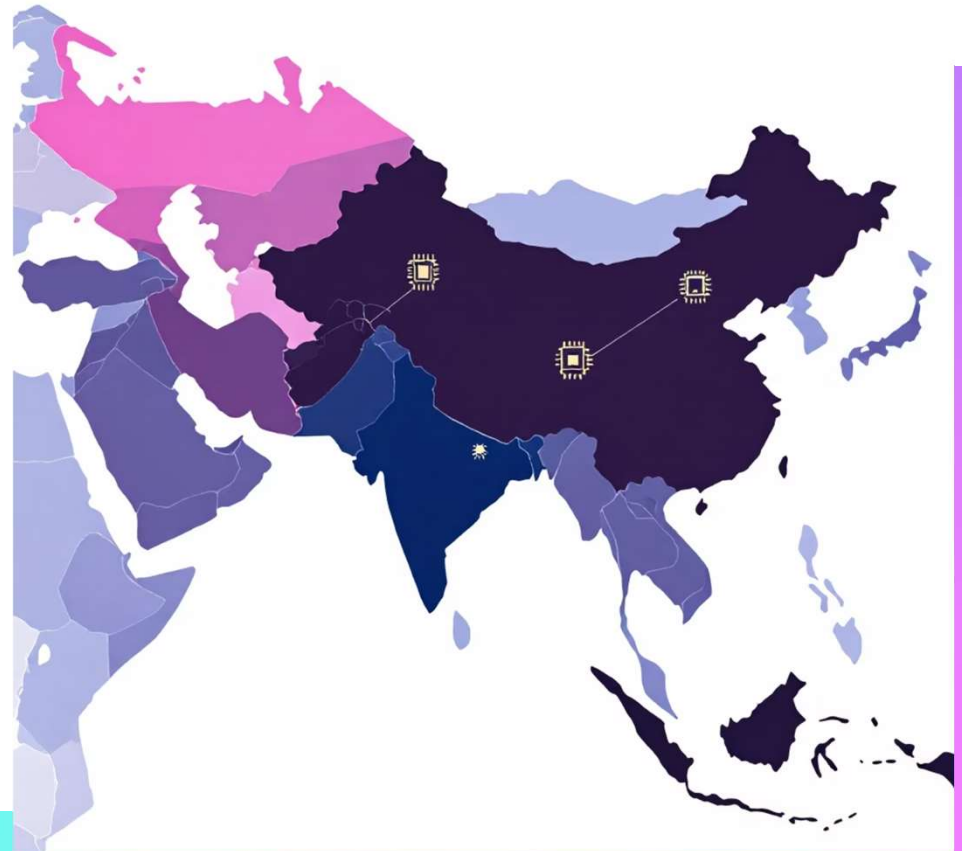
Geopolitical & Industry Dynamics



Regional Adoption Drivers

- **China:** Pursuing domestic RISC-V development to counter trade restrictions
- **India:** "Digital India RISC-V" programme creating semiconductor sovereignty
- **Europe:** European Processor Initiative investing €8 billion in RISC-V capabilities

These initiatives reflect growing concerns about **digital sovereignty** and the need for **supply chain resilience** in the semiconductor industry.



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